

Does the freeze detector work?

FoGNet 1-D CNN · Daphnet (10 Parkinson's patients) · ankle accelerometer

0.71

of real freezes caught
(sensitivity)

0.84

of walking correctly cleared
(specificity)

0.78

balanced accuracy
(sens + spec) / 2

CONFUSION MATRIX · n=8982

true ↓ pred →	no-freeze	freeze
no-freeze	6842 correct walk	1285 false alarm
freeze	248 MISSED freeze	607 caught freeze

VERSUS BASELINES

Approach	Sens	Spec
FoGNet (CNN)	0.71	0.84
Freeze-Index	0.002	0.994
Always 'no-freeze'	0.00	1.00

Each held-out patient is scored once, then pooled
— no patient is in both train and test.

Why not “accuracy”? A model that NEVER calls a freeze scores 90.5% accuracy — yet catches zero freezes and prevents zero falls. A missed freeze can mean a fall, so sensitivity & specificity are the honest metrics here, not accuracy.

DATASET & METHOD

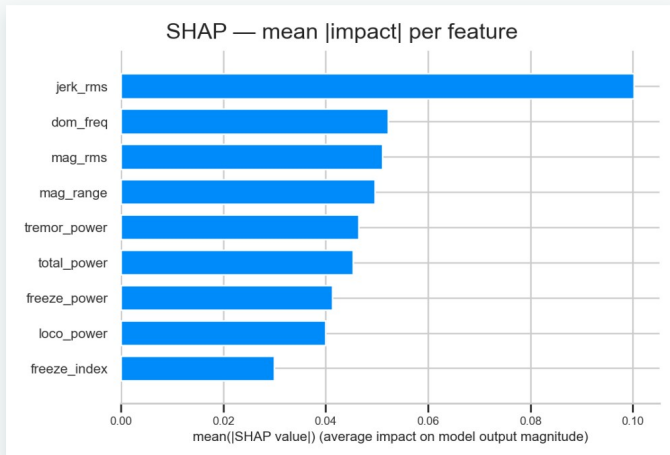
- 8982 four-second windows (2 s hop), one ankle accelerometer @ 64 Hz
- Freeze prevalence 9.5% (855 freeze / 8127 walk) — a rare, safety-critical event
- Leave-One-Subject-Out across all 10 patients; metrics pooled over held-out windows
- S04 & S10 have no freeze episodes → used for training only (sensitivity undefined as a test fold)

Per-patient sensitivity spans 0.45–0.98 (wide between-subject variability is expected for real Parkinson's gait; the pooled 0.71 is the headline).

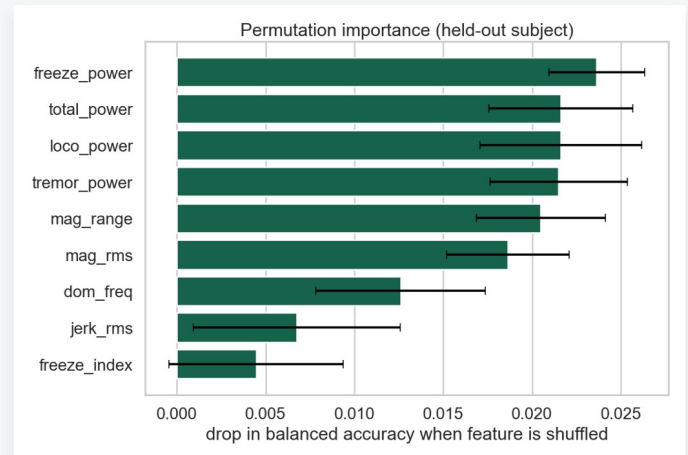
Which signals flag a freeze?

RandomForest over 9 engineered features · held-out patient S05

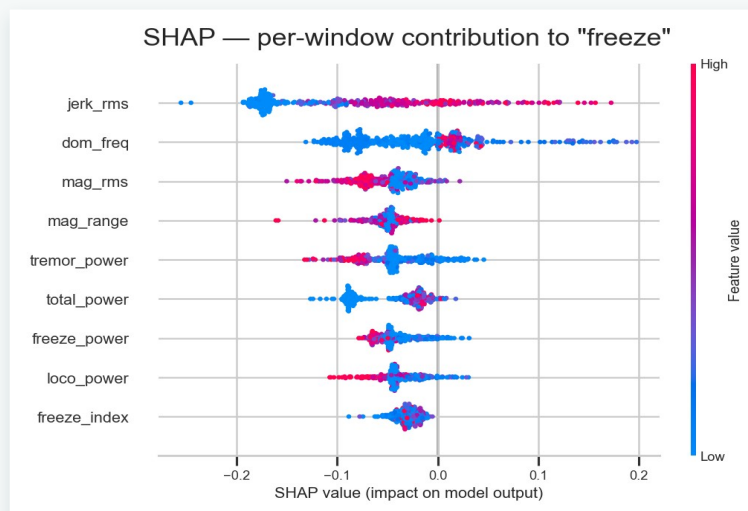
Takeaway: both SHAP and permutation importance rank the classical Freeze-Index **last** of 9 features — so the dead FI baseline is explained, not just observed. Movement dynamics (jerk, dominant frequency, magnitude) carry the signal.



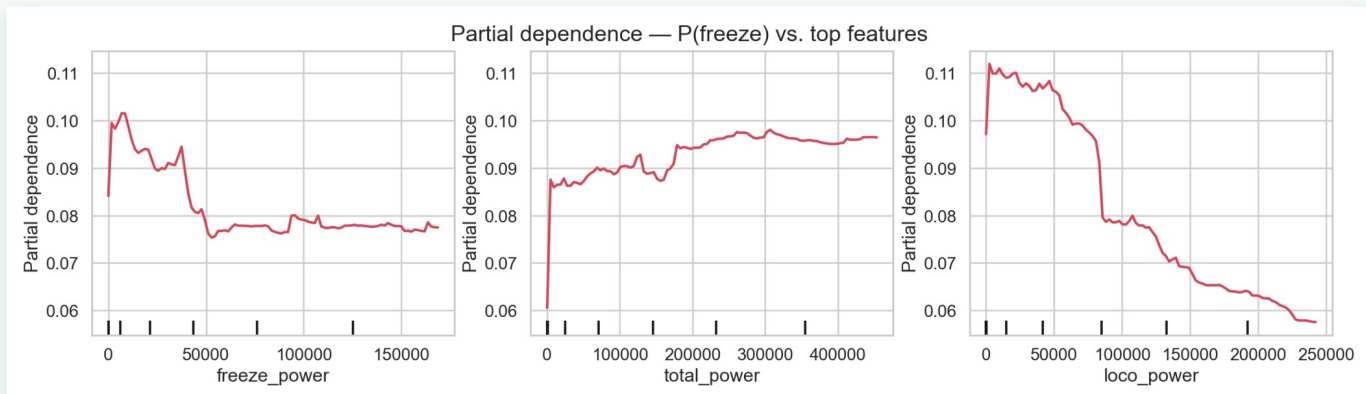
SHAP — mean impact per feature



Permutation importance — accuracy drop



SHAP beeswarm — red = high feature value



Partial dependence — $P(\text{freeze})$ vs. the three most influential features